GATE-AII BRANCHES
Engineering Mathematics



Probability and Statistics

Lecture No.- 15



By- Rahul sir

Recap of Previous Lecture



Topic Concepts of Gaussian distribution

Topics to be Covered



Topic	Gaussian distribution continued
Topic	Question based on uniform distribution and gaussian distribution

Guassian Distribution :

$$\text{Z score} = \frac{X-\mu}{\sigma} = N(\mu, \sigma^2)$$

mean $= \mu$

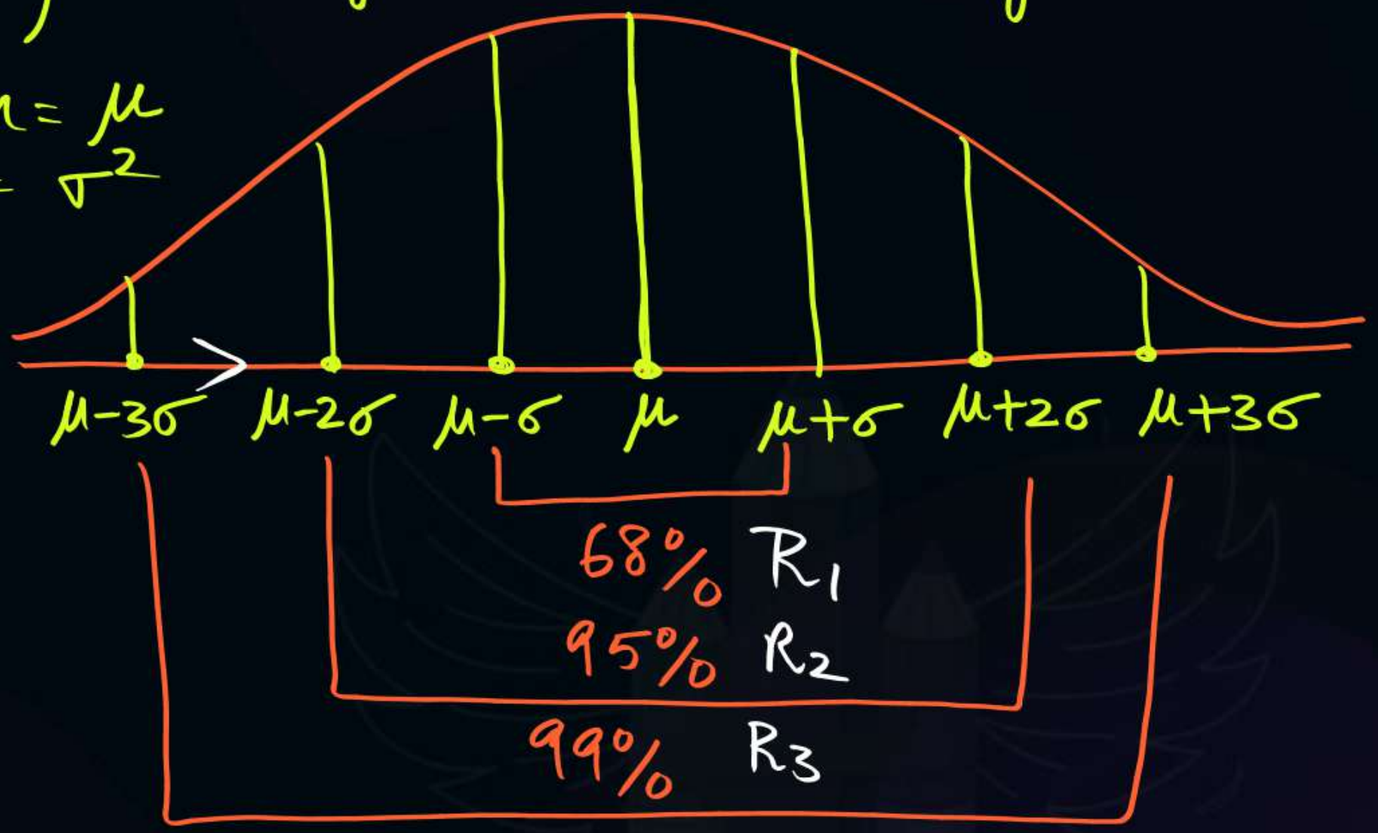
S.D $= \sigma^2$

$$Z = \frac{\text{Random} - \text{mean}}{\text{st deviation}}$$

var

standarize

standraized



$$f(x) = N(\mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$ Gaussian Dist

pdf

$$z = \frac{x-\mu}{\sigma}$$ (statistical Parameter) zscore

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$ (standard Gaussian Distribution

If $\mu = 0$ (zero mean) $\sigma = 1$ (unit variance)

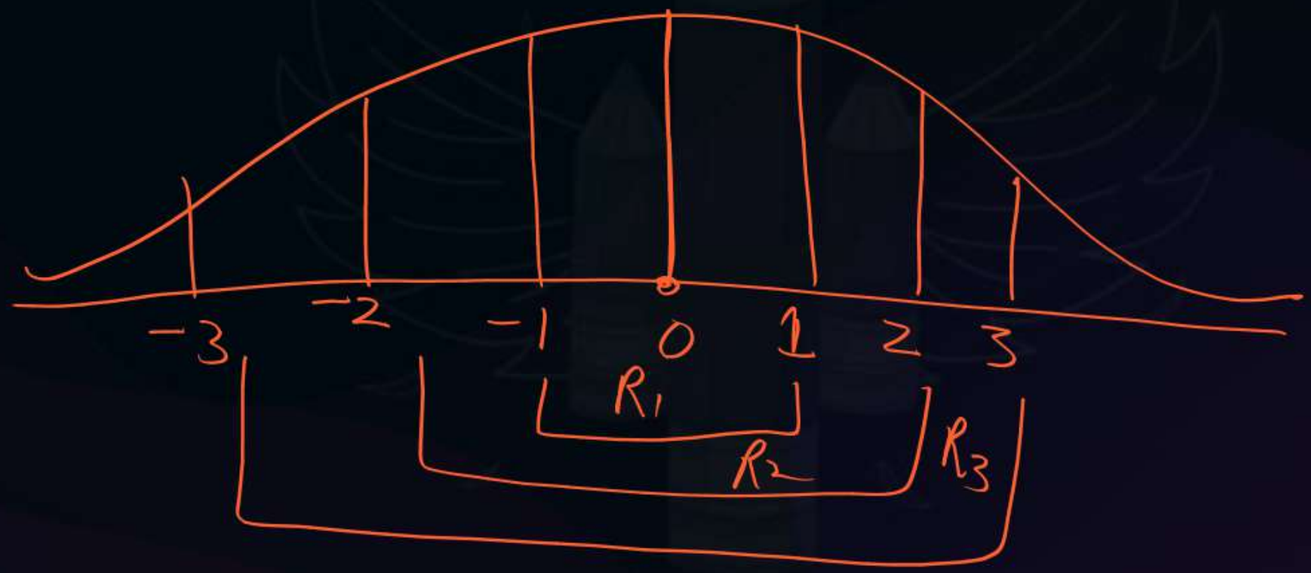


$\mu = 0$
$\sigma = 1$

R_1 region → 68%

R_2 region → 95%

R_3 region → 99.71%



R_1 region : $(-1 \le z \le 1)$

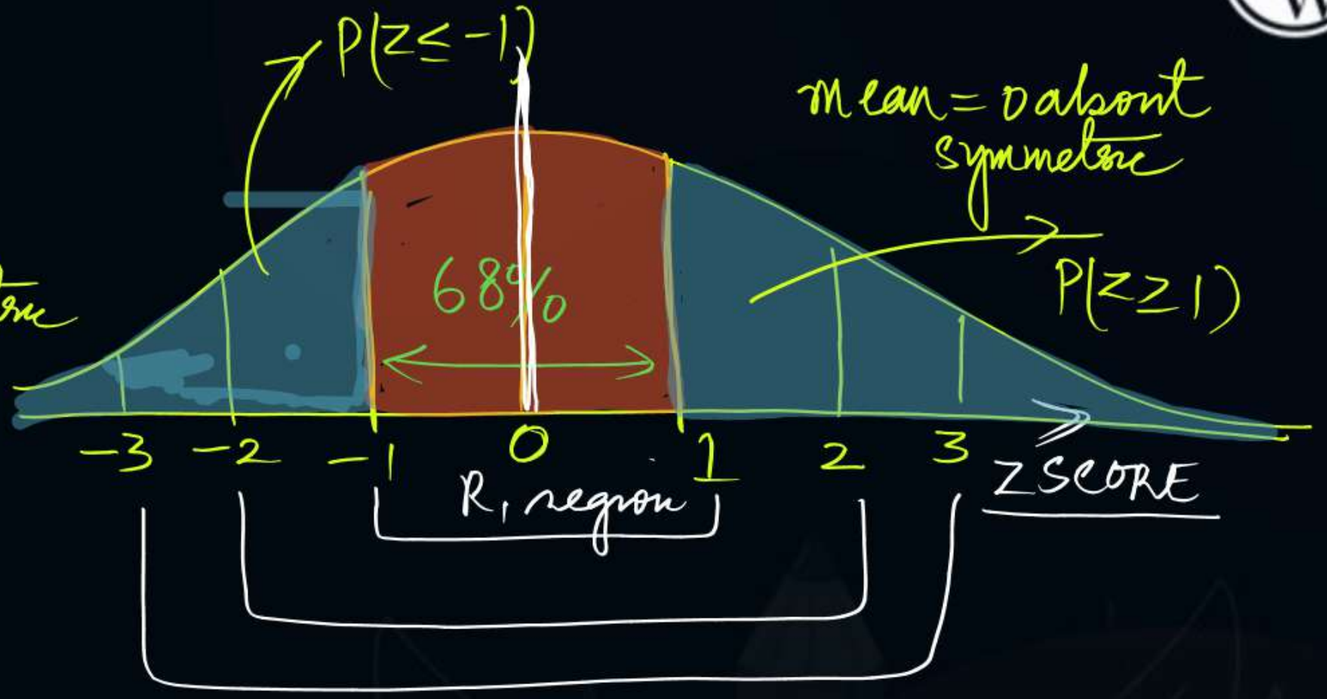
$P(-1 \le z \le 1) = 68\% = 0.6834$

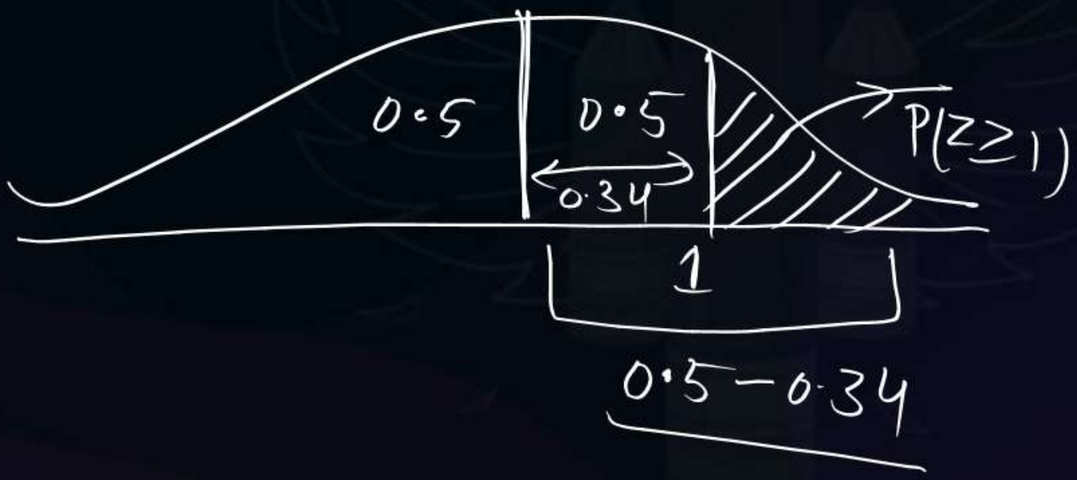
$P(0 \le z \le 1) = 0.3417$ } symmetric

$P(-1 \le z \le 0) = 0.3417$ }

$P(z \ge 1) = P(z \le -1)$

$= 0.5 - P(0 \le z \le 1)$





R_2 region

$P(-2 \le z \le 2) = 0.9545$

$P(0 \le z \le 2) = 0.4772$

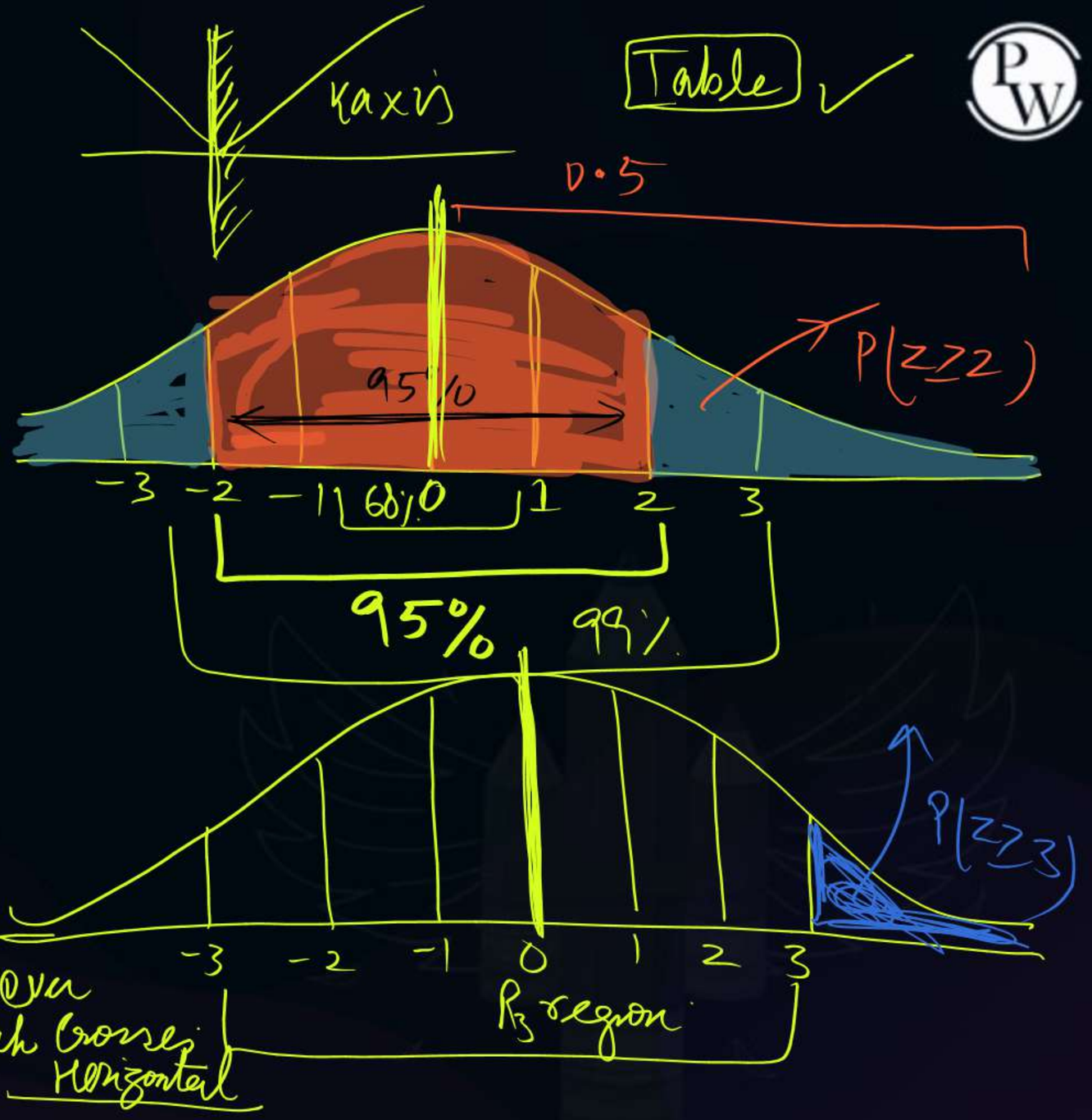
$P(z \ge 2) = 0.5 - P(0 \le z \le 2)$

R_3 region

$P(-3 \le z \le 3) = 0.9971$

$P(0 \le z \le 3) = 0.4985$

$P(z \ge 3) = 0.5 - P(0 \le z \le 3)$



$$\# \quad P(a \le X \le b) = P\left(\frac{a-\mu}{\sigma} \le \frac{X-\mu}{\sigma} \le \frac{b-\mu}{\sigma}\right)$$

X is a Gnassian random variable $N(\mu, \sigma^2)$

$$= P\left(\frac{a-\mu}{\sigma} \le Z \le \frac{b-\mu}{\sigma}\right)$$

$$\# \quad P(X \ge a) = P\left(\frac{X-\mu}{\sigma} \ge \frac{a-\mu}{\sigma}\right)$$

$$= P\left(Z \ge \frac{a-\mu}{\sigma}\right)$$

$$\# \quad P(X \le a) = P\left(\frac{X-\mu}{\sigma} \le \frac{a-\mu}{\sigma}\right) = P\left(Z \le \frac{a-\mu}{\sigma}\right)$$

Topic : Probability and Statistics

$$N(\mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Read The Prob :-
Z-SCORE

#Q. A nationalized bank has found the daily balance available in its saving accounts follows a normal distribution with a mean of Rs. 500 and a standard deviation of Rs. 50. The percentage of saving account holders, who maintain an average daily balanced more than Rs. 500 is-

X = Guassian Distribution

Given Data MEAN = 500 ₹

S.D = 50 ₹

$P(X > 500)$

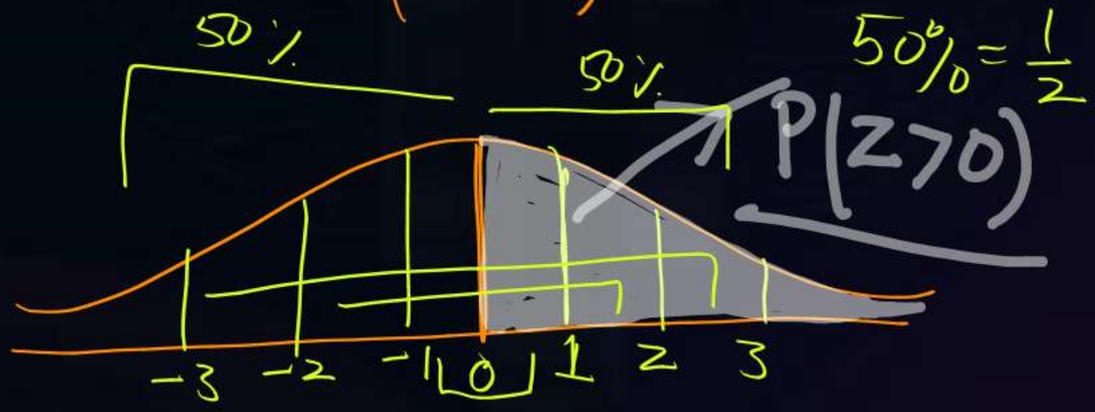
Simple method

$$P(X > 500) = P\left[\frac{X-\mu}{\sigma} > \frac{500-\mu}{\sigma}\right]$$

Using ZSCORE method

$$= P\left(Z > \frac{500-500}{50}\right)$$

$$= P(Z > 0) = \frac{1}{2} = 50\%$$





Topic : Probability and Statistics

number of trials = 180
success - face 4 failure = $\overline{4}$

#Q. A Die is rolled 180 times using Gaussian random variable. Find the Probability that faces 4 will turn up at least 35 times.

$P(X \geq 35)$ = 35 to 180
$= 1 - P(X=1) - \cdots$

Number of Trials = 180 $P(S) = \frac{1}{6}$ $P(F) = \frac{5}{6}$

Bernoulli Trials
Trials fixed
prob. fixed
} Binomial Distribution

If $B(n,p) \xrightarrow{\text{Large No. of Trials}} N(\mu, \sigma^2)$

Using Gaussian

$$P(X \geq 35) = P\left(\frac{X-\mu}{\sigma} \geq \frac{35-\mu}{\sigma}\right)$$

$$= P\left(Z \geq \frac{35-\mu}{\sigma}\right)$$

μ = binomial
σ^2 = s.t = binomial
$B(n,p)$

$$P(X \geq 35) = P\left(\frac{X-\mu}{\sigma} \geq \frac{35-\mu}{\sigma}\right)$$

$$= P\left(Z \geq \frac{35-30}{5}\right) = P(Z \geq 1)$$

$$= 0.5 - P(0 \leq Z \leq 1)$$

$$= 0.5 - 0.3413$$

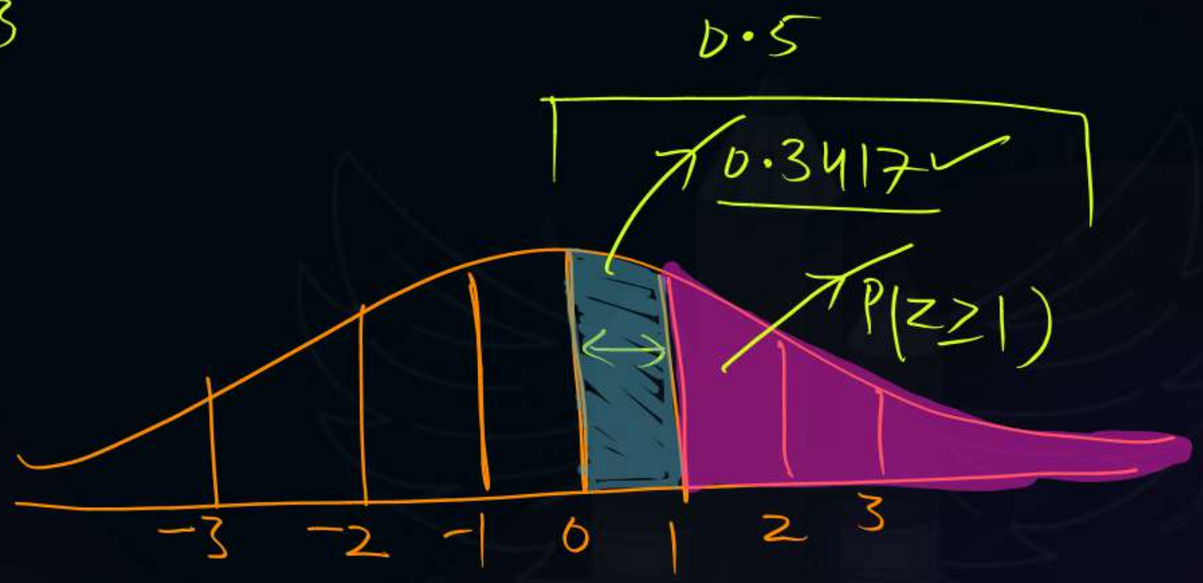
$$= \underline{0.1587}$$

$$\underline{\text{Ans}}$$

$$\mu = np = 180 \times \frac{1}{6} = 30$$

$$\sigma^2 = npq = 180 \times \frac{1}{6} \times \frac{5}{6} = 25$$

$$\sigma = 5$$





Topic : Probability and Statistics

0.3746

#Q. Assume mean height of the soldiers is 68.22 inches with the variance 10.8 inches. How many soldiers in Regiment of 1000 would you expected to be over (6 feet tall). Given that standard Normal Curve X = 0 to 1.15 = 0.3746.

1 Feet = 12 inches
6 feet = 72 inches

Mean = 68.22 inches
Var = 10.8 inches

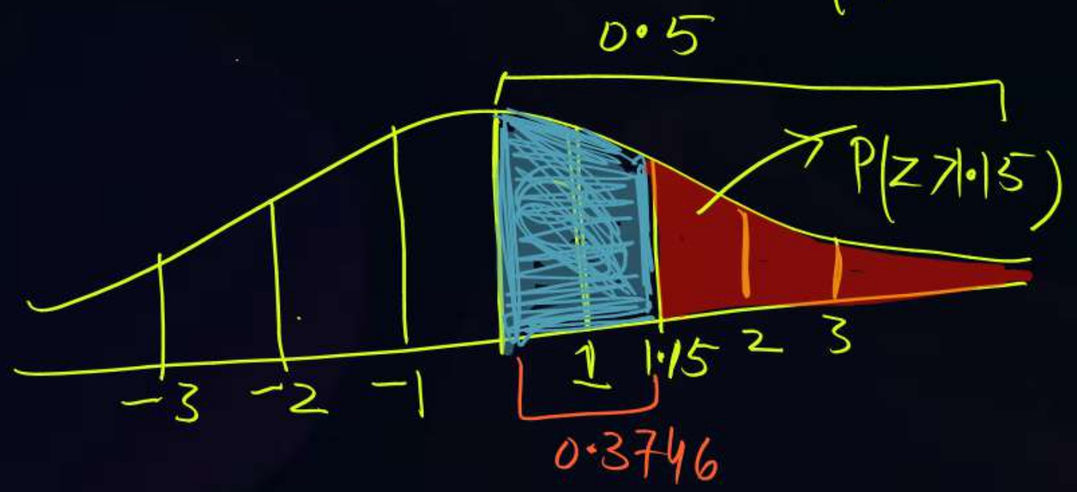
$$P(X > 72 \text{ inches}) = P\left(\frac{x-\mu}{\sigma} > \frac{72-\mu}{\sigma}\right) = P\left(Z > \frac{72-68.22}{\sqrt{10.8}}\right) = 0.1254$$

$$= P(Z > 1.15)$$

$$= 0.5 - P(0 \le Z \le 1.15)$$

$$= 0.5 - 0.3746 = 0.1254$$

$0.1254 \times 1000 = 125$ soldiers (approx)





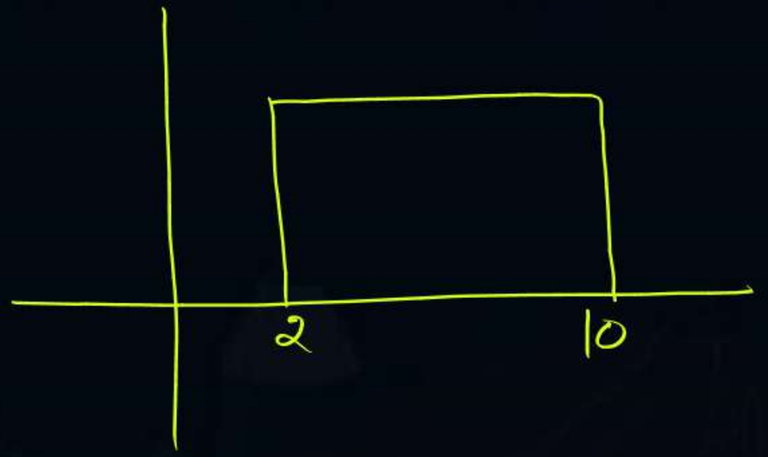
Topic : Probability and Statistics

#Q. A random variable is uniformly distributed over the interval 2 to 10. Its variance will be

$$\text{Variance} = \frac{(b-a)^2}{12}$$

$$= \frac{(10-2)^2}{12}$$

$$= \frac{64}{12} = \frac{16}{3}$$



A. 16/3

B. 6

C. 256/9

D. 36



Topic : Probability and Statistics

#Q. The lengths of a large stock of titanium rods follow a normal distribution with a mean (μ) of 440 mm and a standard deviation (σ) of 1 mm. What is the percentage of rods whose lengths lie between 438 mm and 441 mm?

stock

$\mu = 440$ mm

S.D = 1 mm

$P(438 \le X \le 441) = ?$

$P\left(\frac{438-\mu}{\sigma} \le \frac{X-\mu}{\sigma} \le \frac{441-\mu}{\sigma}\right)$

$= P\left(\frac{438-440}{1} \le Z \le \frac{441-440}{1}\right)$

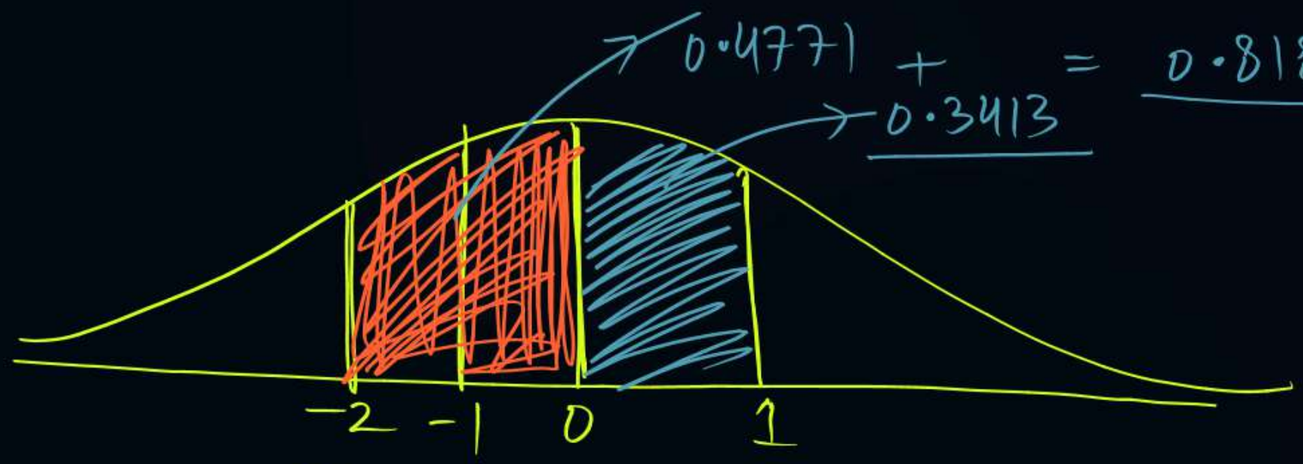
$P(-2 \le Z \le +1)$

A 81.85%

B 68.4%

C 99.75%

D 86.64%



$$= P(-2 \le z \le 0) + P(0 \le z \le 1)$$

$$= 0.4771 + 0.3413$$

$$= 0.8185$$

$$= 81.85\%$$

Topic : Probability and Statistics

$x(-\infty < x < \infty)$
x – Gaussian random var.

#Q. For a random variable $x(-\infty < x < \infty)$ following Normal distribution, the mean is $\mu = 100$. If the probability is $P = \alpha$ for $x \geq 110$. Then the probability x lying between 90 and 110 i.e., $P(90 \leq x \leq 110)$ is equal to

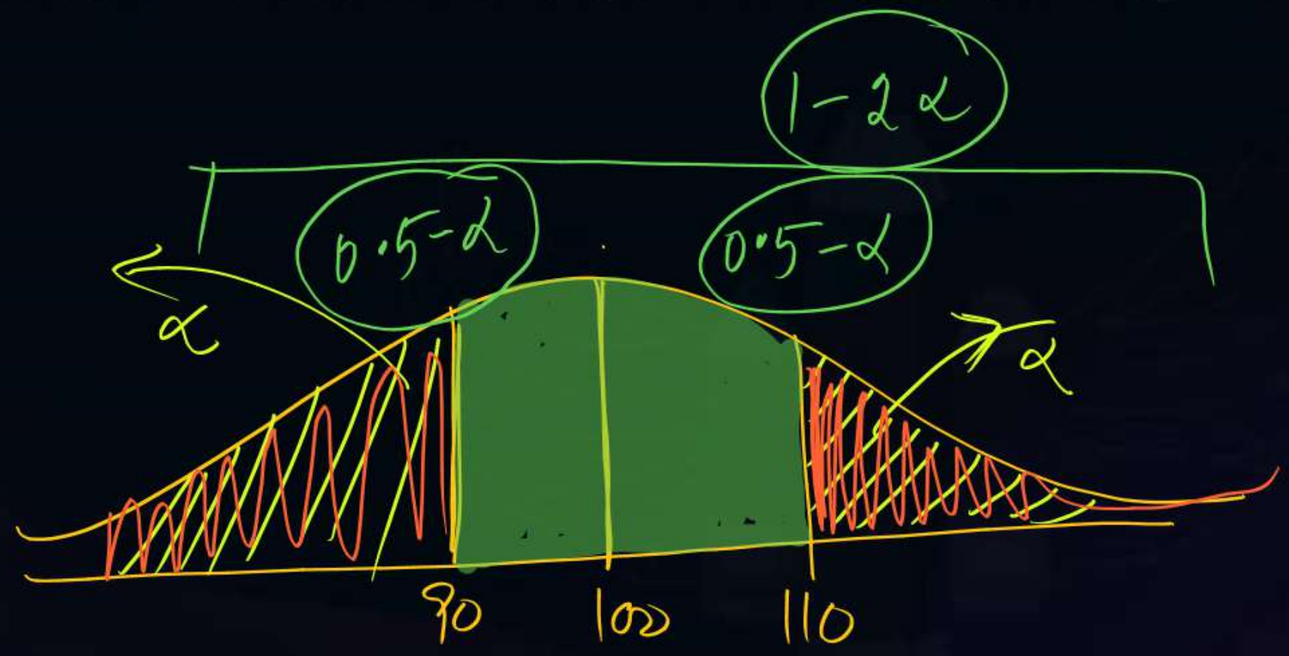
$= 1 - 2\alpha$

A $1 - 2\alpha$

B $1 - \alpha$

C $1 - \alpha/2$

D 2α





Topic : Probability and Statistics

#Q. The annual precipitation data of a city is normally distributed with mean and standard deviation as 1000 mm and 200 mm, respectively. The probability that the annual precipitation will be more than 1200 mm is

mean = 1000 mm

S.D = 200 mm

$$P(X>1200) = P\left(\frac{X-\mu}{\sigma} > \frac{1200-\mu}{\sigma}\right) = P\left(Z > \frac{1200-1000}{200}\right)$$

$$= P\left(Z > \frac{200}{200}\right) = P(Z>1)$$

$= 0.5 - P(0<Z<1)$

$= 0.1587$

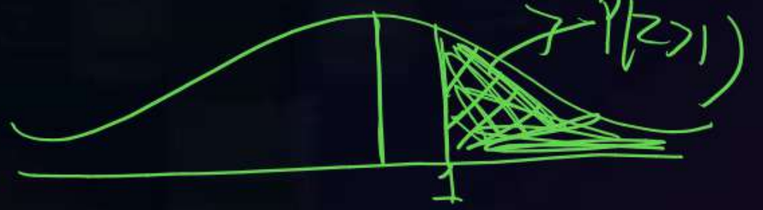
$= < 50\%$

A < 50 %

B 50%

C 75%

D 100%





Topic : Probability and Statistics

$X = N(\mu, \sigma^2)$

$\mu = 1 \quad \sigma^2 = 4 \quad \sigma = \sqrt{4} = 2$

#Q. Let X be a normal random variable with mean 1 and variance 4. The probability $P\{X < 0\}$ is

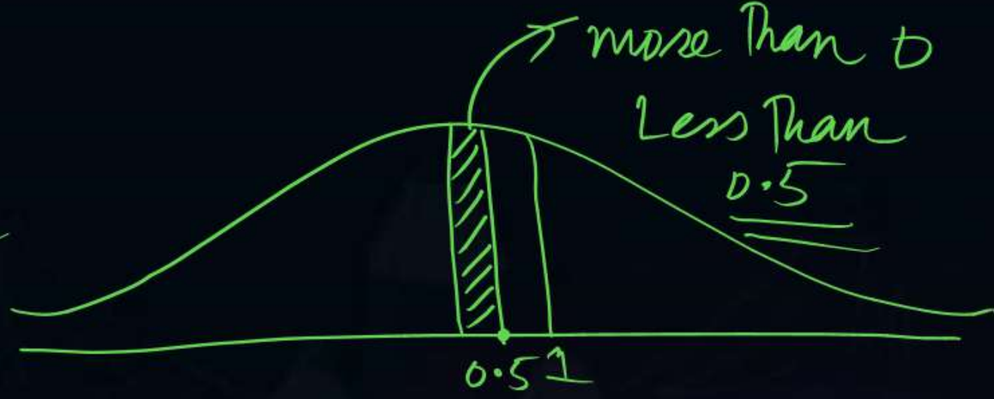
$$P(X<0) = P\left(\frac{X-\mu}{\sigma} < \frac{0-\mu}{\sigma}\right)$$

$$= P\left(Z < \frac{0-1}{2}\right)$$

$$= P(Z < -0.5)$$

because of symmetry

$$= P(Z > 0.5)$$



A 0.5

B Greater than zero less than 0.5

C Greater than 0.5 less than 1.0

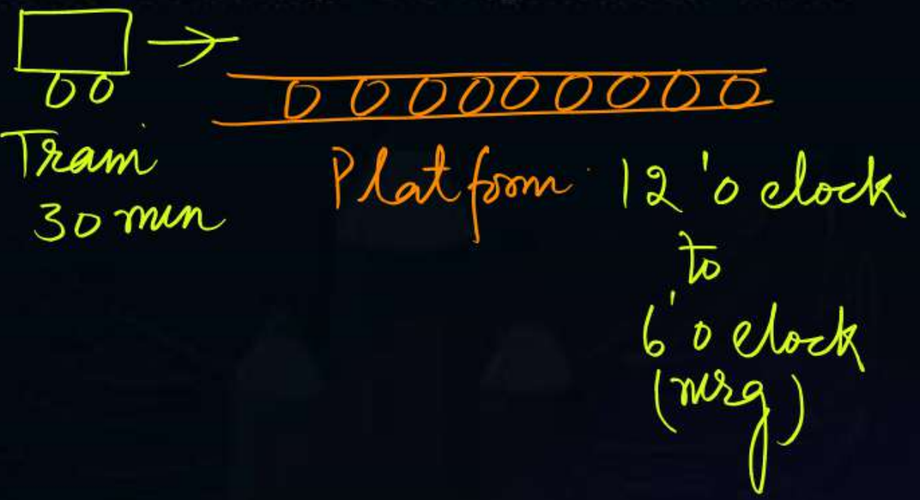
D 1.0



Topic : Probability and Statistics

#Q. Subway trains on a certain line runs every half hour between midnight and six in the morning. What is the probability that the men entering the station at random time during the period will have to wait at least 20 minutes.

$P(X \geq 20 \text{ minutes})$



12:00 to 12:30
12:30 to 1:00
1:00 to 1:30
⋮
5:30 to 6:00

Every 30 minutes ↓ train



Using Uniform distribution $U[0,30]$

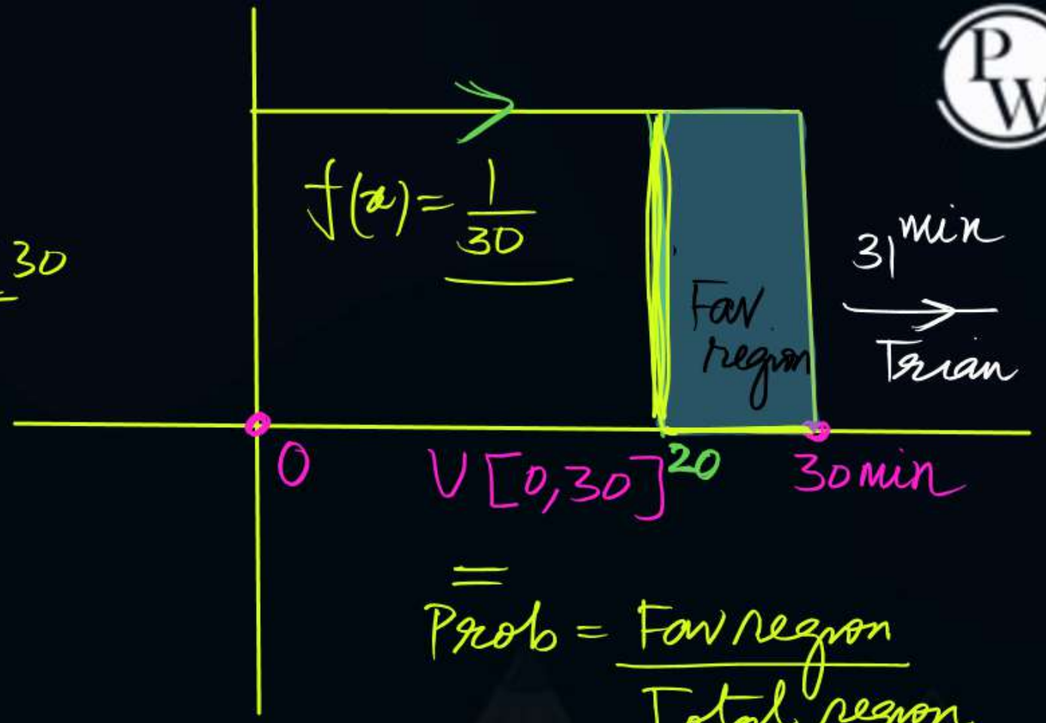
$$f(x) = \frac{1}{(b-a)} = \frac{1}{(30-0)} \quad 0 \le x \le 30$$

$$f(x) = \frac{1}{30}$$

$P(X \ge 20) = P(\text{atleast Twenty min})$

$$P(X \ge 20) = \int_{20}^{30} f(x)\,dx$$

$$= \int_{20}^{30} \frac{1}{30} = \frac{10}{30} = \frac{1}{3}$$



$=$

$$\text{Prob} = \frac{\text{Fav region}}{\text{Total region}}$$

$$= \frac{10}{30} = \frac{1}{3} \text{ Ans}$$



Topic : Probability and Statistics

$P(X \le 0) = (a, b)$

#Q. If X is a uniformly distributed random variable with mean 1 and variance 4/3. Find $P(X \le 0)$

$\text{mean} = 1 \quad \frac{a+b}{2} = 1 \quad a+b = 2 \quad -(1)$

$\text{variance} \quad \frac{(b-a)^2}{12} = \frac{4}{3} \quad (b-a) = \pm 4 \quad -(2)$

$P(X \le 0)$

$a+b=2$
$b-a=4$
$2b=6$
$b=3$
$a+b=2 \quad (a=-1)$
$a+3=2$

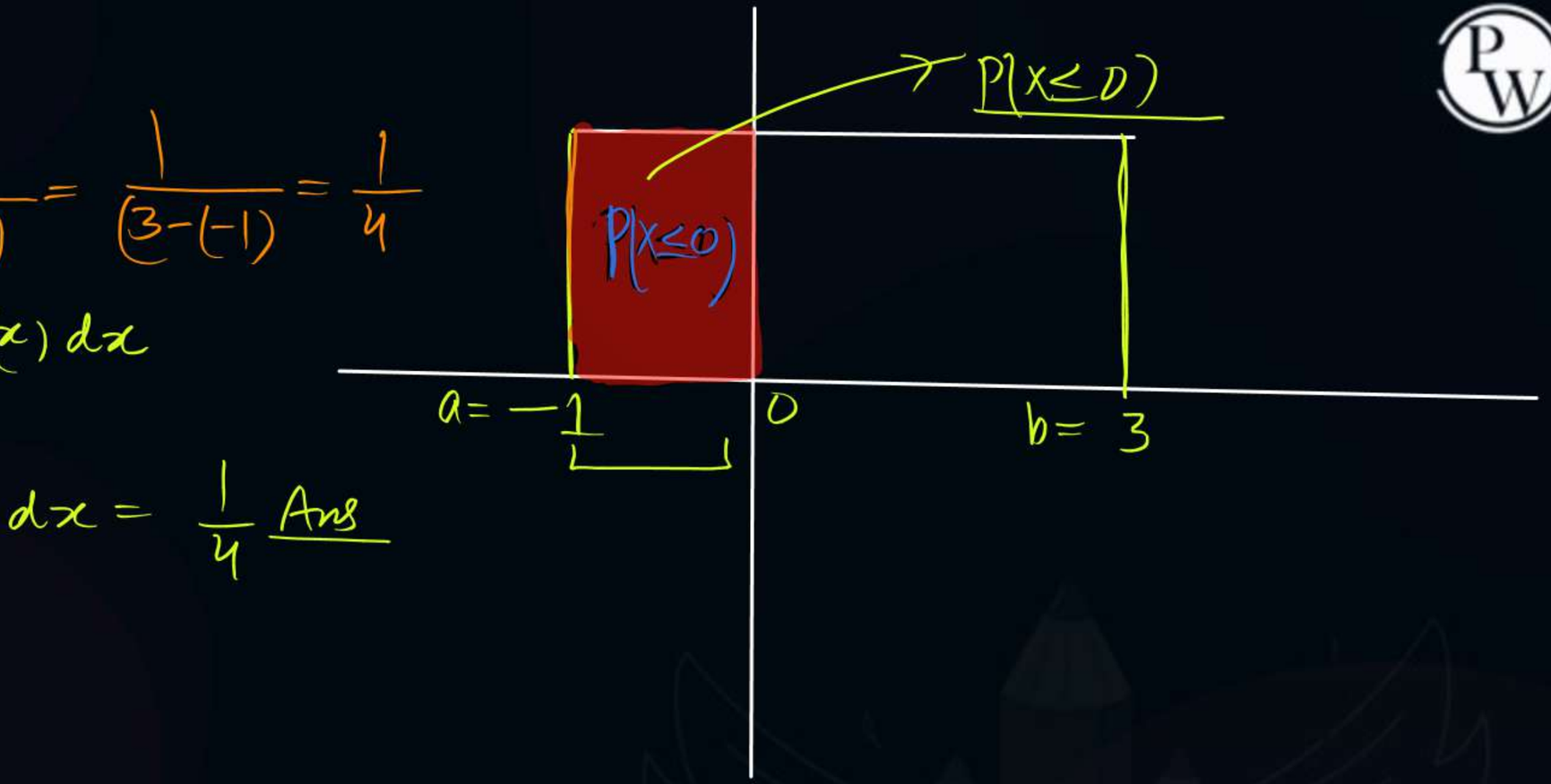
$a+b=2$
$b-a=-4$
$2b=-2$
$b=-1$
$a=3$

$-1 \quad 3$

$$f(x) = \frac{1}{(b-a)} = \frac{1}{(3-(-1))} = \frac{1}{4}$$

$$P(x \le 0) = \int_{-1}^{0} f(x)\,dx$$

$$= \int_{-1}^{0} \frac{1}{4}\,dx = \frac{1}{4} \quad \text{Ans}$$



2 days – revise
Rahul Sir PW

THANK - YOU